Plant growth and development

Introduction and Growth

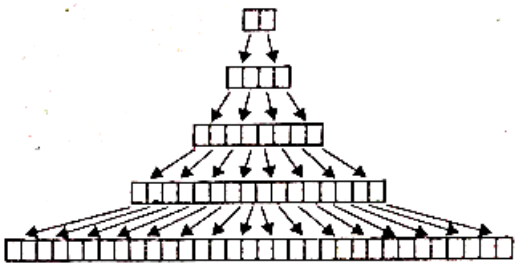
1. Primary growth occurs in plants due to
1) Apical meristem
2) Intercalary meristem
3) Lateral meristem
4) Both (1) and (2)

2. The period of growth is generally divided into three phases, namely, meristematic, elongation and maturation. Identify the regions on the basis of following features:
A. Increased vacuolation and new cell wall deposition.
B. Cells are rich in protoplasm, large conspicuous nucleus, thin cellulosic cell wall with plasmodesmata connections.
C. Cells attain their maximum sizes in terms of wall thickening and protoplasmic modification.
1) A—Maturation; B—Elongation; C-Meristematic
2) A-Elongation; B-Meristematic; C-Maturation
3) A-Elongation; B-Maturation; C-Meristematic
4) A-Maturation; B--Meristematic; C-Elongation

3. Read the following statements, and select the correct option.
A. Growth is accompanied by metabolic processes that occur at the expense of energy.
B. When the rate of catabolism exceeds the rate of anabolism, growth occurs.
1) Only (A) is incorrect.
2) Only (B) is incorrect.
3) Both (A) and (B) are incorrect.
4) Both (A) and (B) are correct.

4. After meiotic division in a cell, if one daughter cell continues to divide, while the other differentiate and matures, then this type of growth is mathematically expressed as
1) $W_1 = W_0 e^{rt}$
2) $L_0 = L_t + rt$
3) $W_0 = W_1 e^{rt}$
4) $L_t = L_0 + rt$

5. The given image represents



1) Arithmetic growth as shown during differentiation of embryo
2) Geometric growth as shown by a root elongating at a constant rate
3) Geometric growth as shown during the initial phase of embryogenesis
4) It will be represented by a straight line when plotted against time

6. Find the incorrect statement with respect to growth
1) Plant growth is always determinate.
2) It is the most fundamental and conspicuous characteristic of a living being.
3) Growth is measurable.
4) In open form of growth, new cells are always being added.

7. Growth can be measured in various ways. Which of these as be used as parameters to measure growth?
1) Increase in cell number
2) Increase in cell size
3) Increase in surface area
4) All of the above

8. Growth at the cellular level is principally a consequence of increases in the amount of
1) Protoplasm
2) Cell wall only
3) Apoplast
4) None of the above

Differentiation, Dedifferentiation, Redifferentiation and Development

9. Plasticity in plant growth means that
1) Plant roots are extensible
2) Plant development is dependent on the environment
3) Stems can extend
4) None of the above

10. Plant follows different pathways in response to environment or phases of life to form different kinds of structures. This ability is called
1) Expansion
2) Plasticity
3) Differentiation
4) Senescence

11. Development is controlled by

1) Phytochromes only
2) Intrinsic factor only
3) Intercellular, genetic and environmental factors
4) Intracellular factors only

12. Growth and differentiation in plants is
1) Closed and open
2) Open and open
3) Open and closed
4) Closed and closed

13. In which of the following plants, the leaves of the juvenile plant are different in shape from those in mature plants?
1) Coriander and larkspur
2) Cotton and buttercup
3) Buttercup and larkspur
4) All of the above

14. Read the given assertion and reason and select the correct option.
Assertion: During differentiation, cells undergo few to major structural changes both in their cell wall and protoplasm.
Reason: To form a tracheary element, the cells lose their protoplasm, develop a very strong, elastic, lignocellulosic secondary cell walls, to carry water to long distances even under extreme tension.
1) If both assertion and reason are true and reason is the correct explanation of the assertion
2) If both assertion and reason are true but reason is not the correct explanation of the assertion
3) If assertion is true but reason is false
4) If both assertion and reason are false

15. Development is considered as the sum of
1) Growth and senescence
2) Expansion and cell division
3) Growth and differentiation
4) Growth and cell division

16. How many of the following are the products of redifferentiation?

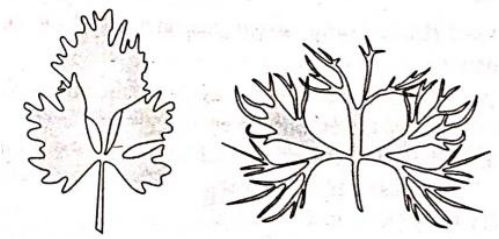
phellogen, phellem, phelloderm, interfascicular cambium, secondary xylem, secondary phloem, primary xylem, procambium, complementary cells

1) Three
2) Five
3) Four
4) Six

17. Read the following statements and choose the incorrect option
1) The phenomenon of heterophylly is an example of plasticity.

2) Differentiation is considered as the sum of growth and development.
3) Development in plants is under the control of intrinsic and extrinsic factors.
4) Light, temperature, water, oxygen, nutrition, etc. are the extrinsic factors that control the development in plants.

18. Which of the given condition is exhibited by the diagram given below?



1) Developmental heterophylly
2) Environmental heterophylly
3) Differentiation
4) Redifferentiation

19. All changes that an organism goes through during its life cycle is termed as
1) Development
2) Growth
3) Differentiation
4) Dedifferentiation

20. Plant hormone that can be used to increase length of grape stalk and improvement in the shape of fruit is
1) Cytokinin
2) Auxin
3) Gibberellin
4) ABA

21. Choose the correct option for auxin.
1) It is synthesized generally at root and shoot apices.
2) It is largely a growth inhibitor.
3) Its existence was reported by Miller.
4) It cannot be used as herbicide.

22. Which of the following is chemically identical to ABA?
1) Abscission II
2) Dormin
3) Inhibitor B
4) All of the above

23. Ethylene is used for
1) Retarding ripening of tomatoes
2) Hastening of ripening of fruits
3) Slowing down ripening of apples
4) Both (2) and (3)

24. Natural cytokinins are synthesized in
1) Root apices

2) Developing shoot buds
3) Young fruits
4) All of the above

25. Gaseous PGR is
1) Acetylene
2) Ethylene
3) Methane
4) Benzaldehyde

26. Coconut water contains
1) ABA
2) Auxin
3) Cytokinin
4) Gibberellin

27. Refer to the given statements.
Statement I: Axillary buds, in actively growing herbaceous plants, generally remain dormant.
Statement II: This is due to apical dominance which is under the influence of auxins.
1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true and statement II is false.
4) Statement I is false and statement II is true.

28. Choose the odd one out with respect to the function of gibberellins
1) Hastens maturity period in juvenile conifers
2) Increase in length of grape stalk
3) Internodal elongation just before flowering in rosette plants
4) Apical hook formation in dicot seedling

29. Which one is the precursor of ethylene?
1) Methionine
2) Tryptophan
3) Xanthophyll
4) Acetyl Co-A

30. Read the following statements.
a) Cytokinin is primarily concerned with cell division.
b) Auxin is synergistic to cytokinin w.r.t morphogenesis.
c) Ethylene breaks seed and bud dormancy.
d) Abscisic acid stimulates the stomatal closure.
How many of the above given statements are correct?
1) Two
2) One
3) Four
4) Three

31. Discovery of first cytokinin is from the autoclaved DNA of a
1) Plant
2) Human
3) Fish
4) Endosperm cell

32. Find the correct match.
1) Auxin-F. W, Went
2) GA-Miller et. al.
3) ABA – Antiageing hormone
4) Ethylene-Carotene derivative

33. The effect of apical dominance can be overcome by which of the following hormone?
1) IAA
2) Ethylene
3) Gibberellin
4) Cytokinin

34. How many of the following statements are correct regarding the plant hormone ethylene?
(a) Can cause internode and petiole elongation in deep water rice plants
(b) Is involved in ripening of climacteric fruit
(c) Induces flowering in mango
(d) Causes bolting in rosette plants
(e) Induces seed dormancy in peanut
1) Two
2) Three
3) Four
4) Five

35. The hormone that was first isolated from human urine was
1) Auxin
2) Gibberellin
3) Cytokinin
4) Abscisic acid

36. What will happen if the terminal bud of a plant is removed?
1) The plant will die
2) The lateral buds will grow
3) The shoot will die
4) All of its leaves will fall

37. Cytokinins help produce all except
1) New leaves
2) Chloroplast in leaves
3) Rooting in stem cutting
4) Lateral shoot growth and adventitious shoot formation

38. Auxins that have been isolated from plants are
1) IAA and NAA
2) IAA and IBA
3) NAA and 2,4-D
4) All of the above

39. Read the following statements:
(i) It is used to prepare weed-free lawn.
(ii) It promotes flowering in pineapple.
The above functions are carried out by
1) Auxin
2) Gibberellin
3) Cytokinin
4) ABA

40. Find the incorrectly matched pair.
1) Auxin-Promotes flowering in pineapple
2) Gibberellins-Foolish seedling disease of rice
3) Ethylene-Promotes female flowers in cucumber
4) Cytokinin-Promotes apical dominance

41. Refer to the given statements.
Statement I: Gibberellins enhance the longitudinal growth of intact plants.
Statement II: Gibberellin is called plant growth promoter.
1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true and statement II is false.
4) Statement I is false and statement II is true

42. A PGR that can fit in both the categories, that is, growth promoters and inhibitors, is
1) Abscisic acid
2) Ethylene
3) Kinetin
4) Dormin

43. Apical bud inhibits the growth of lateral buds, This effect can be counteracted by
1) Auxin application
2) Spraying auxin in root
3) The application of 2,4-D
4) Cytokinin application

44. Bioassay of auxin is
1) Avena curvature test
2) Triple action test
3) Tobacco pith culture test
4) Dwarf maize test

45. The hormone which was discovered through foolish seeding disease' of rice is

1) Indole 3-acetic acid
2) Ethylene
3) Gibberellin
4) Kinetin

46. Refer to the given statements.

Statement I: Both natural and synthetic auxins have been used extensively in agricultural and horticultural practices.

Statement II: Auxin is widely applied in tea plantations and hedge-making.

1) Both statement I and statement II are true.
2) Both statement I and statement II are false
3) Statement I is true and statement II is false.
4) Statement I is false and statement II is true

47. The most widely used PGR in agriculture

1) Promotes abscission in leaves and flowers
2) Initiates germination in peanut seeds
3) Increases the rate of respiration
4) All of the above

48. Find the correct application of auxin.

1) Acts as anti-transpirant and induces closure of stomata
2) Promotes flowering in long day plants
3) Increases sugarcane yield up to 20 tonnes per acre
4) Help to initiate rooting in stem cuttings

49. To increase sugar production in sugarcanes, they are sprayed with

1) IAA
2) Cytokinin
3) Gibberellin
4) Ethylene

Answer Keys

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
4	2	2	4	3	1	4	1	2	2	3	2	1	2	3
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
2	2	2	1	3	1	4	2	4	2	3	1	4	1	3
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
3	1	4	2	1	2	3	2	1	4	1	2	4	1	3
46	**47**	**48**	**49**											
1	4	4	3											